

Exponential, Step, and Impulse Signals

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Key Concepts

1) Exponential signals model decay or growth.

    a) Continuous-time exponential $x(t) = Ae^{-bt}$ decays with increasing time for $b > 0$ and grows for $b < 0$.

    b) Discrete-time exponential $x[n] = A\alpha^n$ decays with increasing time for $0 < \alpha < 1$ and grows for $\alpha > 1$.

2) Exponentially damped sinusoids oscillate at the sinusoid frequency, but have an envelope that decays or grows exponentially.

3) Step signals represent a sudden change from one steady condition to another.

4) Continuous-time step signals are discontinuous at time $t = 0$.

5) Impulse signals represent a very short duration perturbation.

6) Continuous-time impulses $\delta(t)$ are generalized functions.

    a) $\delta(t)$ is zero everywhere except $t = 0$ and has unit area.

b) The sifting property states that the integral of a signal times a shifted impulse is equal to the value of the signal at the time the impulse occurred.

$$\int_{-\infty}^{\infty} f(t)\delta(t - t_0)dt = f(t_0)$$

Exponential, Step, and Impulse Signals

Why? Useful Models

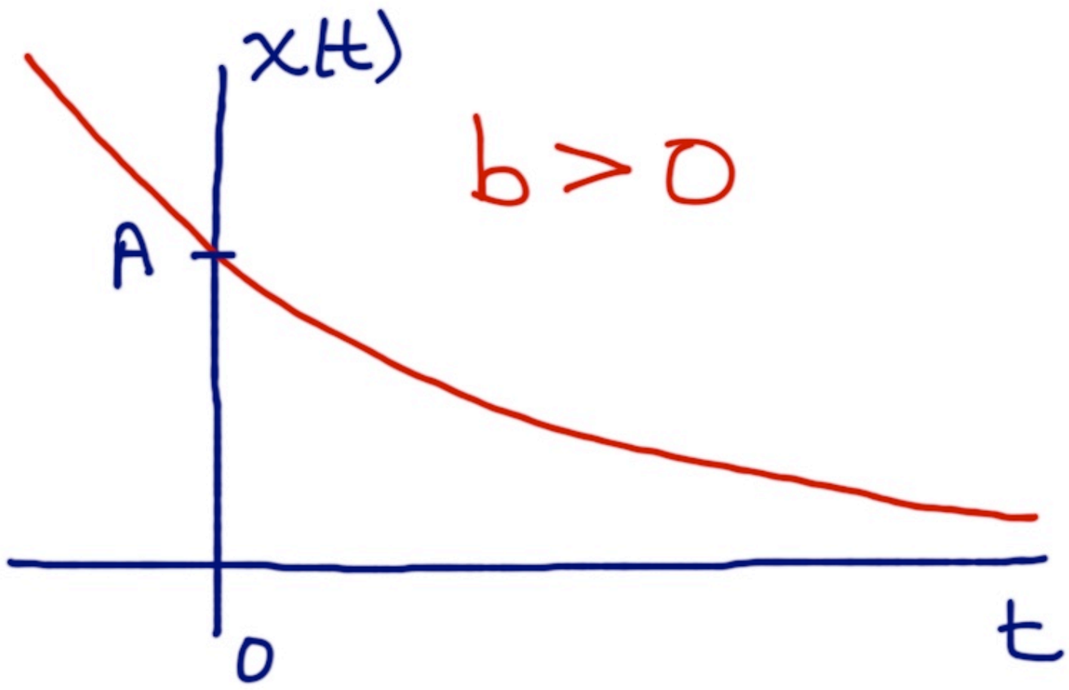
1. Exponentials - decay or growth
2. Steps - sudden change, switch
3. Impulses - perturbation, “kick the tires”

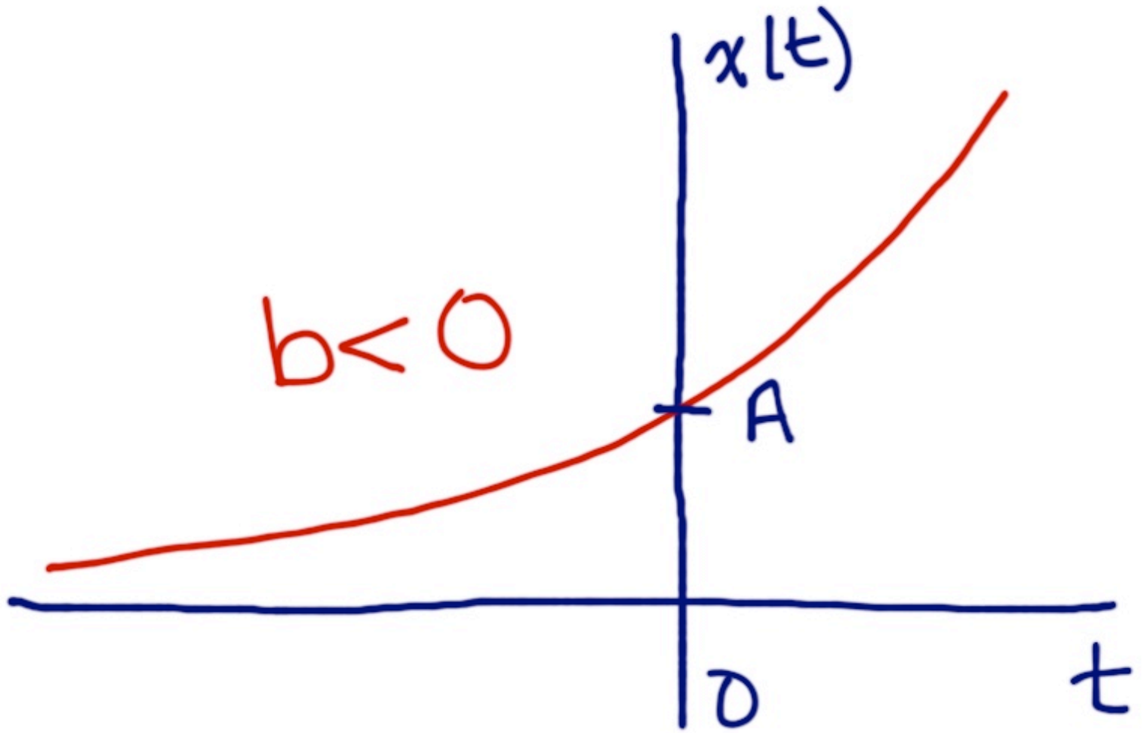
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Continuous-Time Exponentials

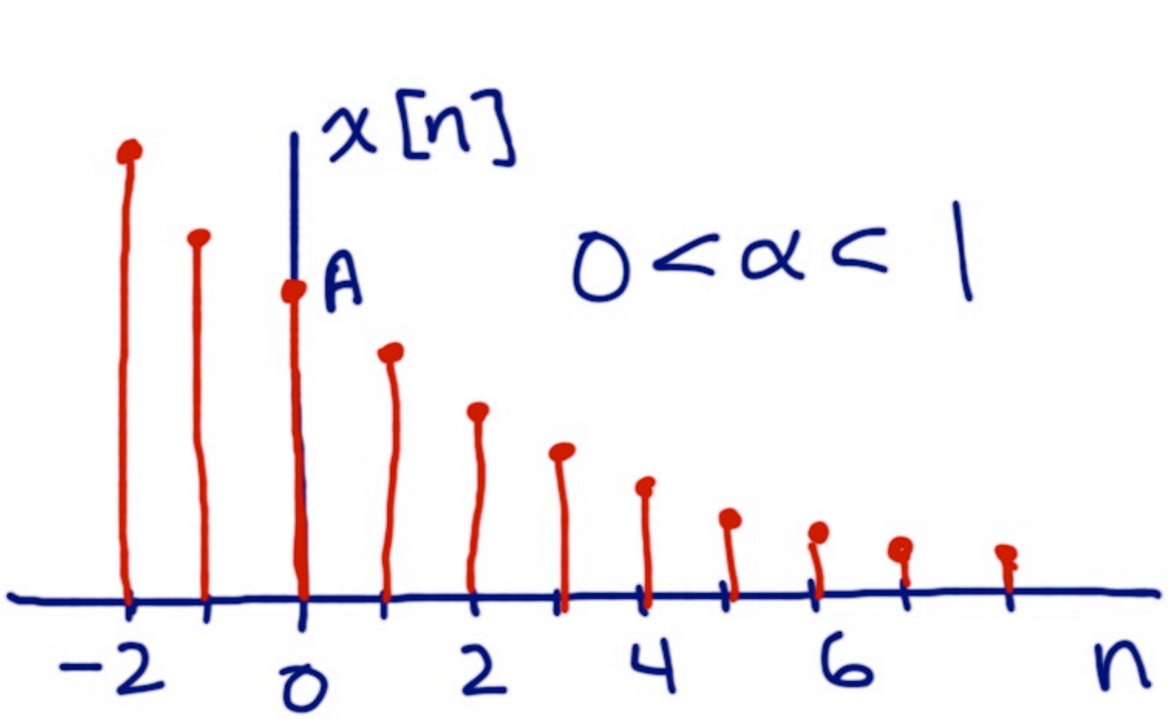
$$x(t) = Ae^{-bt} \quad \text{or} \quad x(t) = A\exp\{-bt\}$$

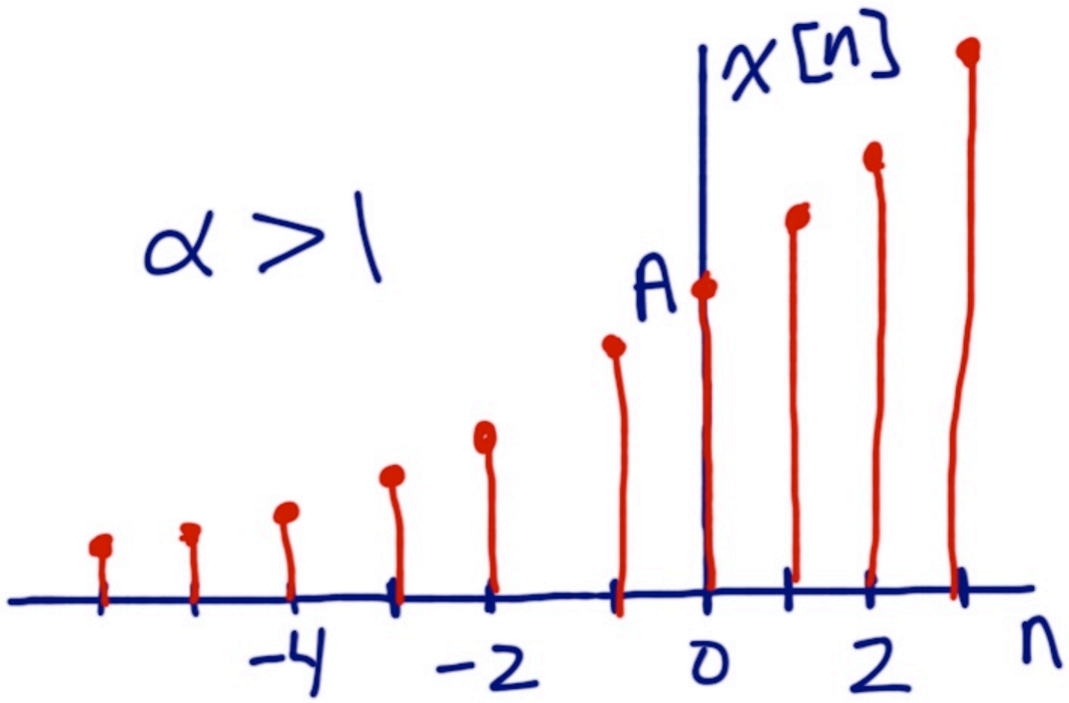




Discrete-Time Exponentials

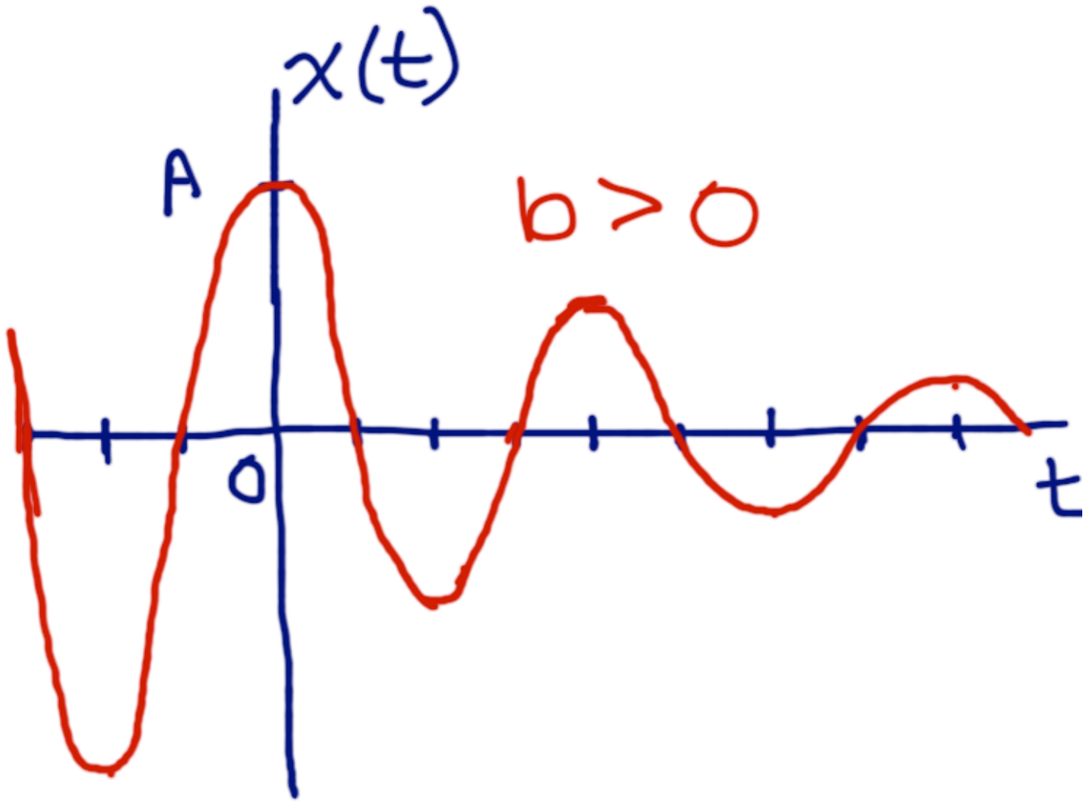
$$\begin{aligned} x[n] &= Ae^{-bn}; \quad \text{let} \;\; \alpha = e^{-b} \\ &= A\alpha^n \end{aligned}$$



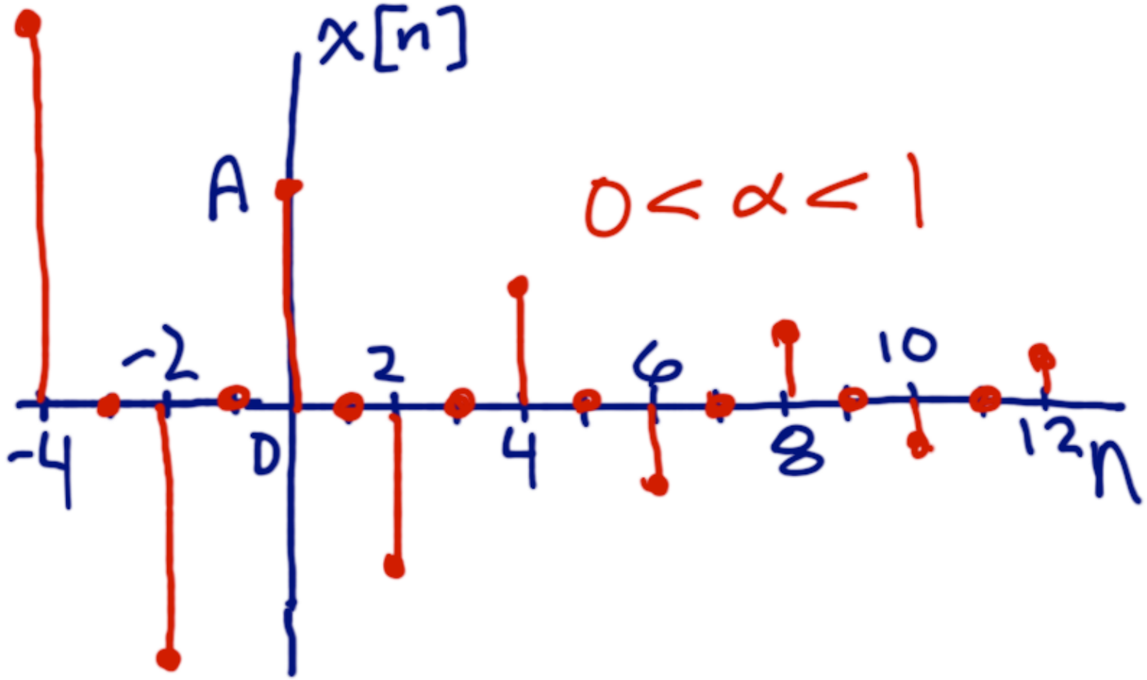


Exponentially Damped Sinusoids

$$x(t) = Ae^{-bt}\cos(\Omega_0 t)$$

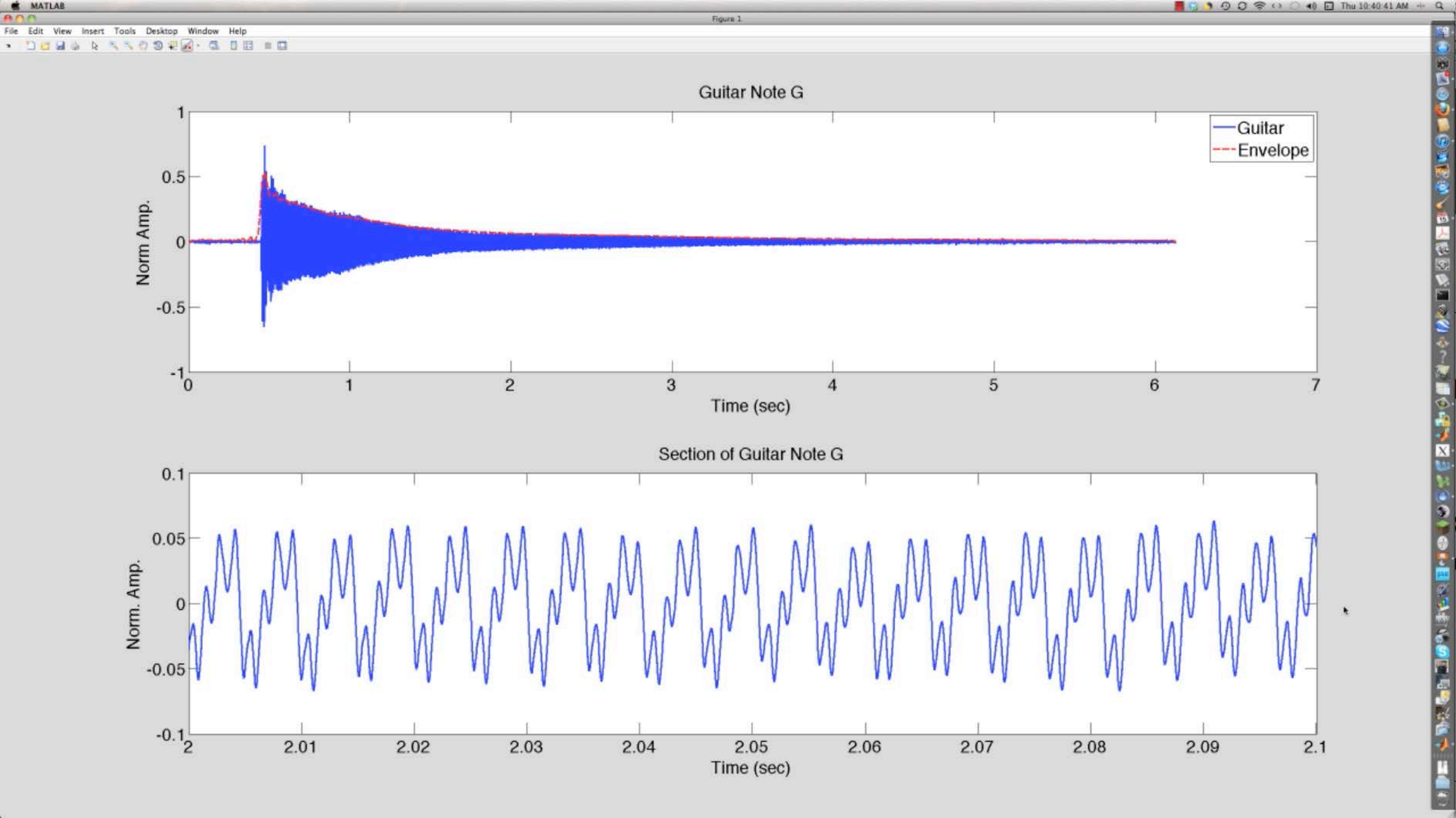


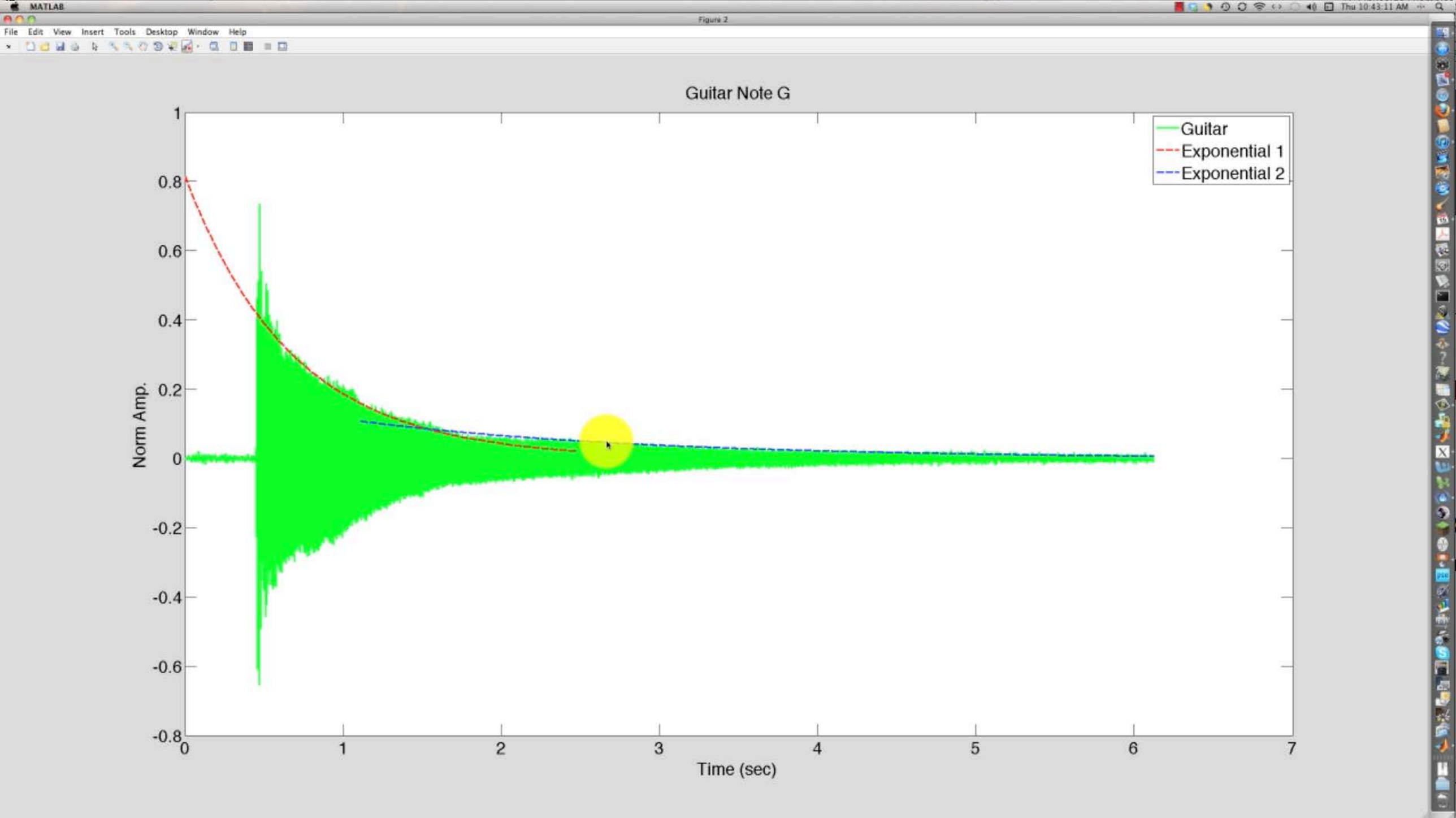
$$x[n] = A\alpha^n \cos(\omega_0 n)$$

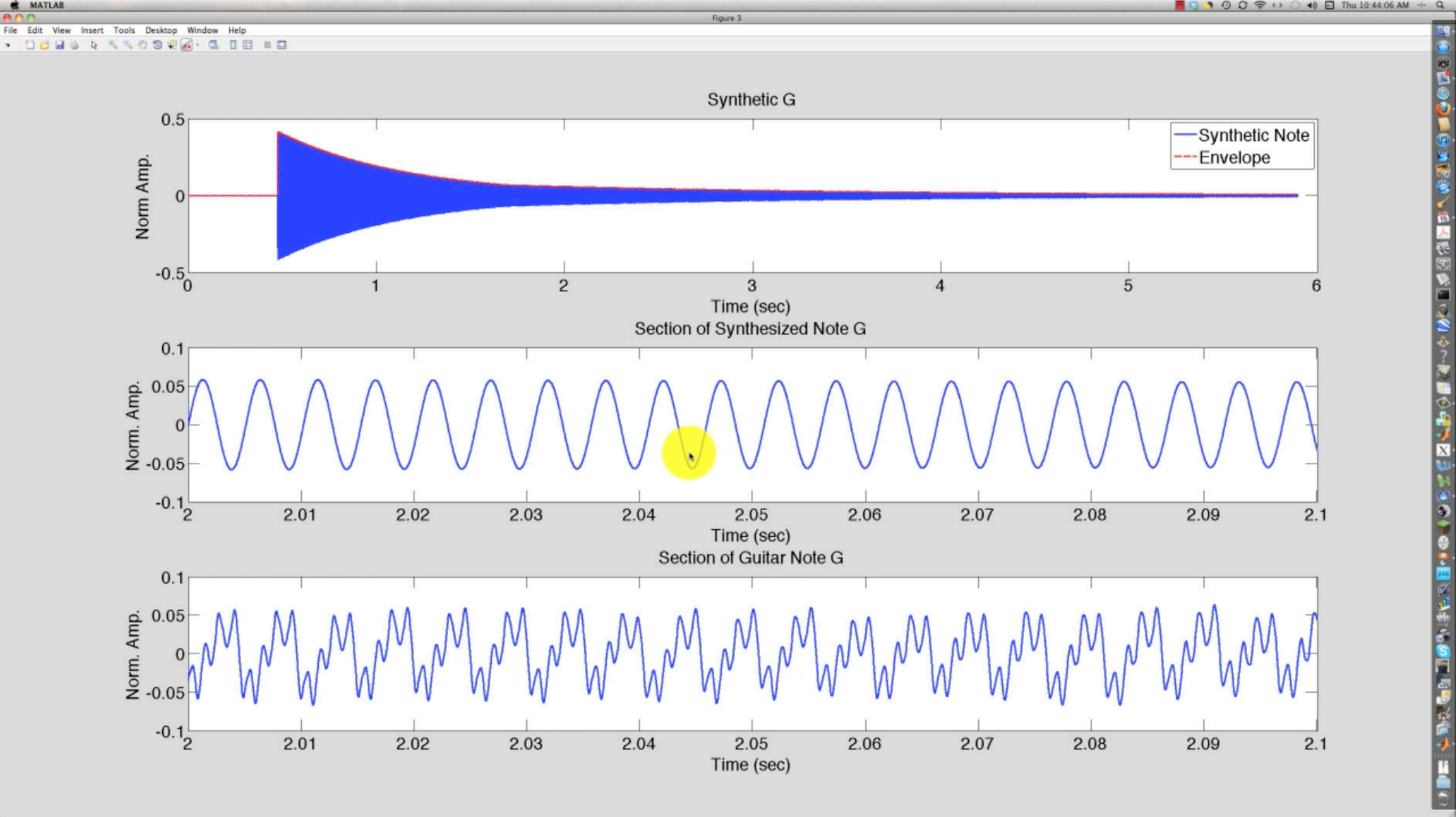


Example

- Pluck guitar string
  1. Envelope of decay
  2. Exponentially damped sinusoid approximation


MATLAB
Figure 1
File Edit View Insert Tools Desktop Window Help
Guitar Note G
Guitar
Envelope
Norm Amp.
Time (sec)
Section of Guitar Note G
Norm. Amp.
Time (sec)

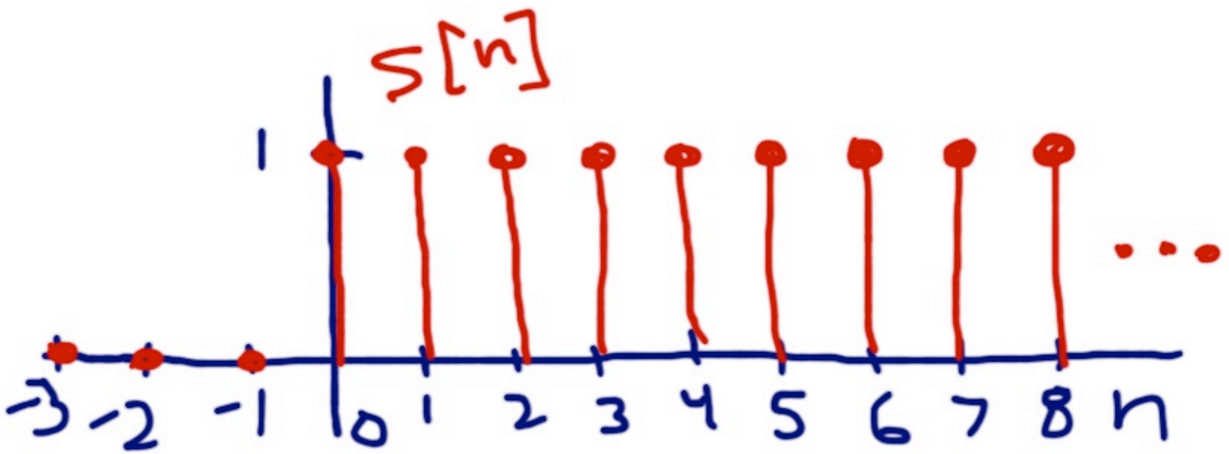

MATLAB
Figure 2
File Edit View Insert Tools Desktop Window Help
Guitar Note G
Guitar
Exponential 1
Exponential 2
Norm Amp.
Time (sec)
1
0.8
0.6
0.4
0.2
0
-0.2
-0.4
-0.6
-0.8
0
1
2
3
4
5
6
7

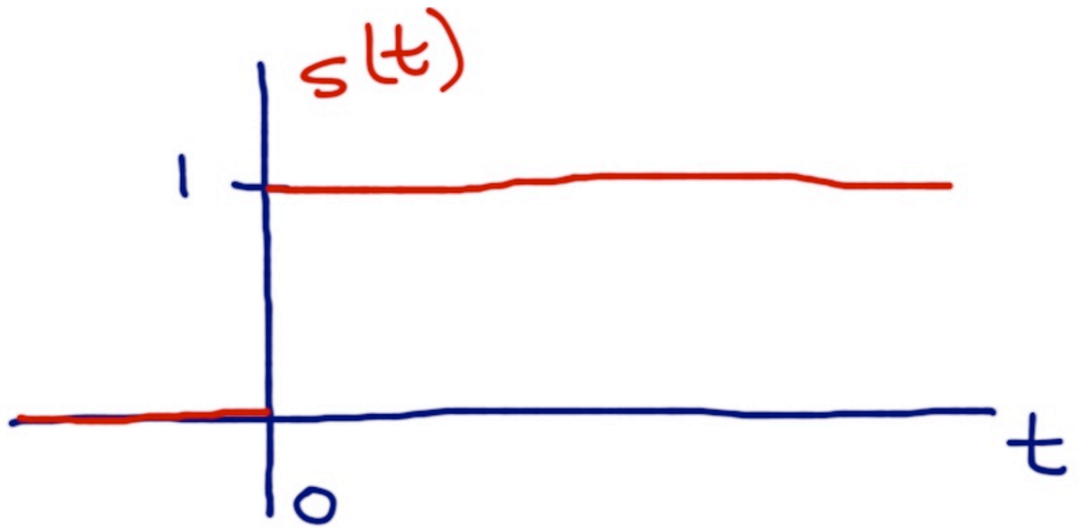

MATLAB
Figure 3
File Edit View Insert Tools Desktop Window Help
Synthetic G
Norm Amp.
Synthetic Note
Envelope
Time (sec)
Section of Synthesized Note G
Norm. Amp.
Time (sec)
Section of Guitar Note G
Norm. Amp.
Time (sec)

Steps

$$s[n] = \begin{cases} 0, & n < 0 \\ 1, & n \geq 0 \end{cases}$$

$$s(t) = \begin{cases} 0, & t < 0 \\ 1, & t > 0 \end{cases}$$



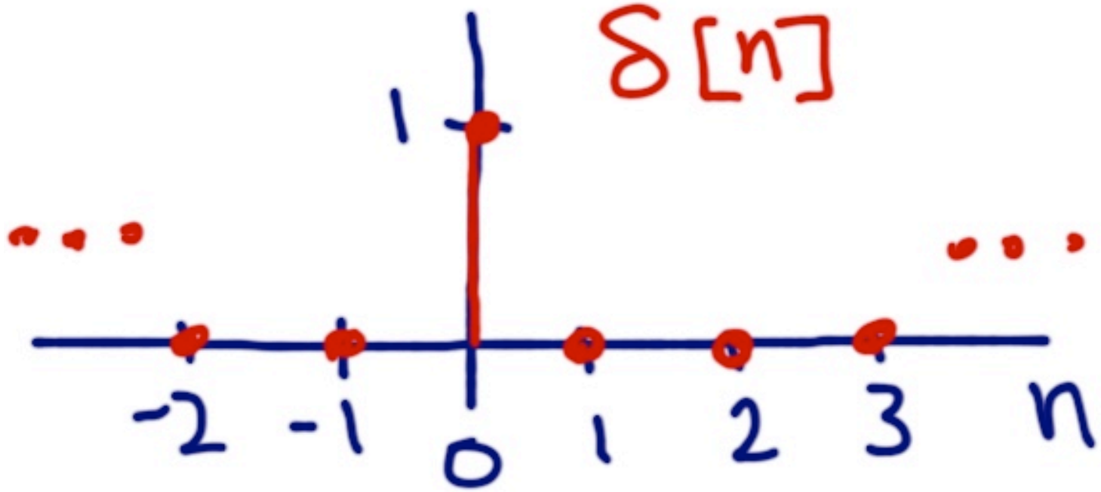


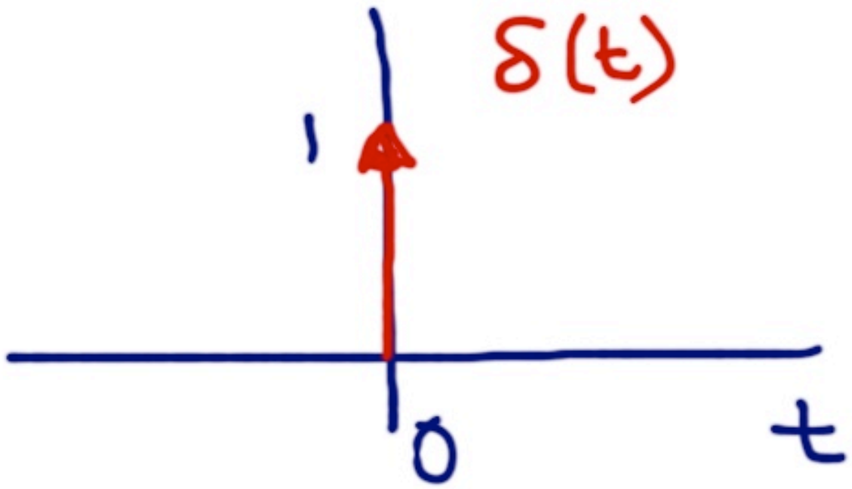
Impulses

$$\delta[n] = \begin{cases} 0, & n \neq 0 \\ 1, & n = 0 \end{cases}$$

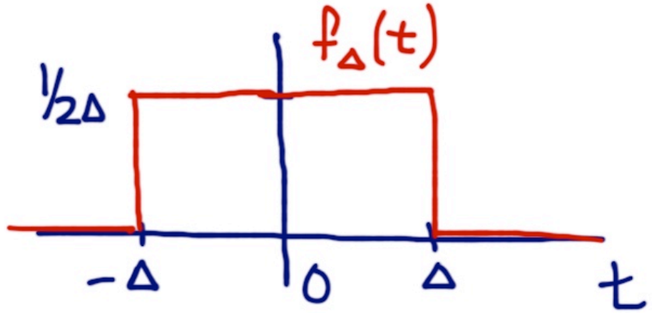
$$\delta(t) = 0, \quad t \neq 0$$

$$\int_{-\infty}^{\infty} \delta(t)dt = 1$$

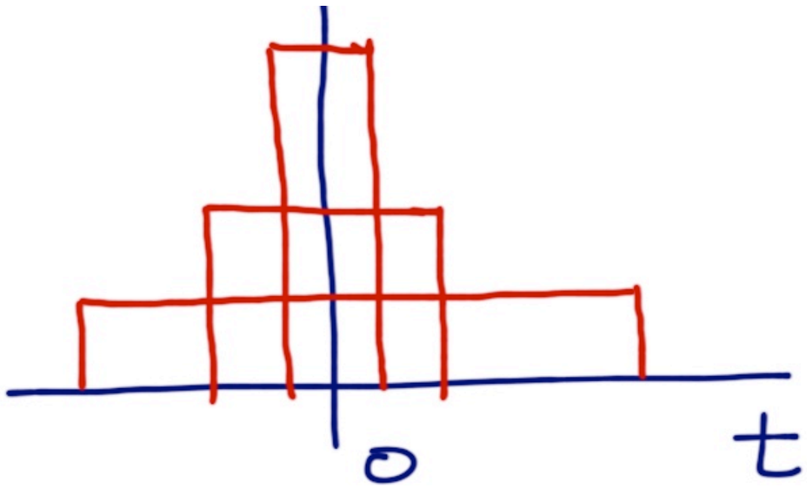




Continuous-Time Impulses



$$\delta(t) \sim \lim_{\Delta \to 0} f_\Delta(t)$$



- Generalized functions
- Only meaningful mathematically within an integral
- Sifting property

$$\int_{-\infty}^{\infty} f(t)\delta(t - t_0)dt = f(t_0)$$

Summary

- Exponentials: natural growth/decay
- Exponentially damped sinusoids: growing/decaying oscillatory behavior
- Steps: switching behavior
- Impulses: unit perturbation

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